

Networking Lab 001

VLAN Segmentation & Switching Fundamentals

Lab Information

Project ID: NET-001

Category: Networking / Switching

Environment: Cisco Packet Tracer Simulation

Network Device: Cisco Catalyst 2960 Switch

Date: 10/06/2026

1. Purpose

The purpose of this networking lab is to demonstrate the configuration and verification of VLAN segmentation using a simulated enterprise switching environment.

This project demonstrates practical knowledge of:

- VLAN creation and management
 - Switch port configuration
 - Network segmentation
 - Access port assignment
 - Basic network troubleshooting
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2. Scenario

A company requires network separation between different departments to improve organization, security, and network management.

The network was segmented into separate VLANs:

VLAN ID	Department
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VLAN 10	HR
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VLAN 20	IT
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3. Environment

Tools Used:

- Cisco Packet Tracer
- Cisco Catalyst 2960 Switch
- Virtual End Devices

Network Components:

- 1 Cisco 2960 Switch
 - 2 Virtual PCs
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4. Network DesignTopology

PC-HR

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Fa0/1

|

Cisco 2960 Switch

|

Fa0/2

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PC-IT

5. Configuration Steps

VLAN Creation

Created department-based VLANs:

- VLAN 10 — HR
 - VLAN 20 — IT
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Switch Port Assignment

Configured switch access ports:

Port	Device	VLAN
Fa0/1	PC-HR	VLAN 10
Fa0/2	PC-IT	VLAN 20

6. Testing & Validation

VLAN Verification

Used switch commands to verify VLAN configuration:

show vlan brief

Confirmed:

- VLAN 10 assigned to HR
 - VLAN 20 assigned to IT
 - Correct switch port assignments
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Connectivity Testing

Performed connectivity testing between VLANs.

Test: PC-HR → PC-IT

Result: ❌ Communication unsuccessful

Explanation: The communication failure was expected because devices in separate VLANs require a Layer 3 routing device to communicate.

7. Evidence Documentation

Evidence 1 — VLAN Creation Verification

Screenshot:

Switch CLI showing VLAN configuration.

The screenshot displays a network simulation environment. In the background, a logical topology shows two PCs, PC-PT/PC-HR and PC-PT/PC-IT, connected to a central 2960-24TT Switch0. The foreground features a terminal window titled 'Switch0' showing the CLI configuration for VLANs. The configuration includes enabling the switch, entering configuration mode, and creating VLANs 10, 20, 1001, 1004, 1005, and 1008. A table at the bottom of the CLI output lists the VLANs and their associated ports.

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/17, Fa0/18, Fa0/19, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 HR	active	
20 IT	active	
1001 dmz-default	active	
1003 token-ring-default	active	
1004 fdxline-default	active	
1005 trinet-default	active	
1008	active	

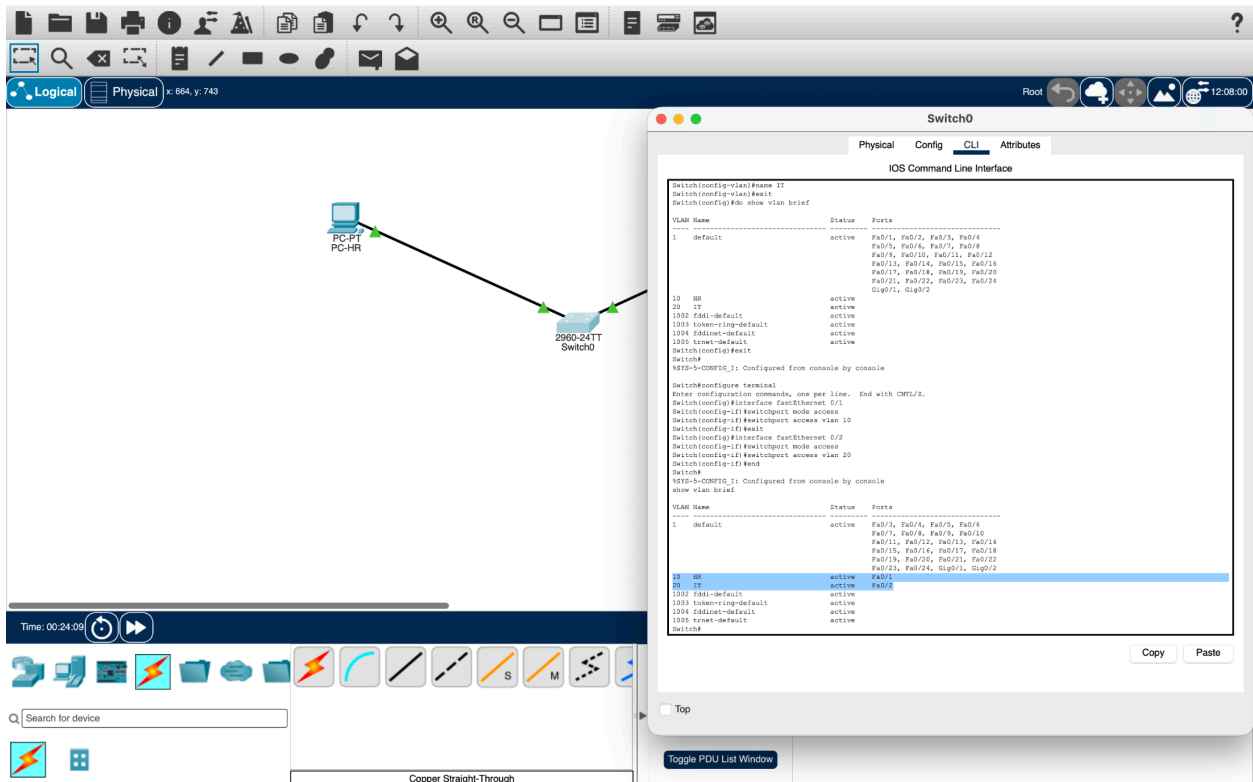
Purpose:

Demonstrates successful creation of logical network segments within the switch.

Evidence 2 — Switch Port VLAN Assignment

Screenshot:

Switch VLAN table showing Fa0/1 and Fa0/2 assignments.



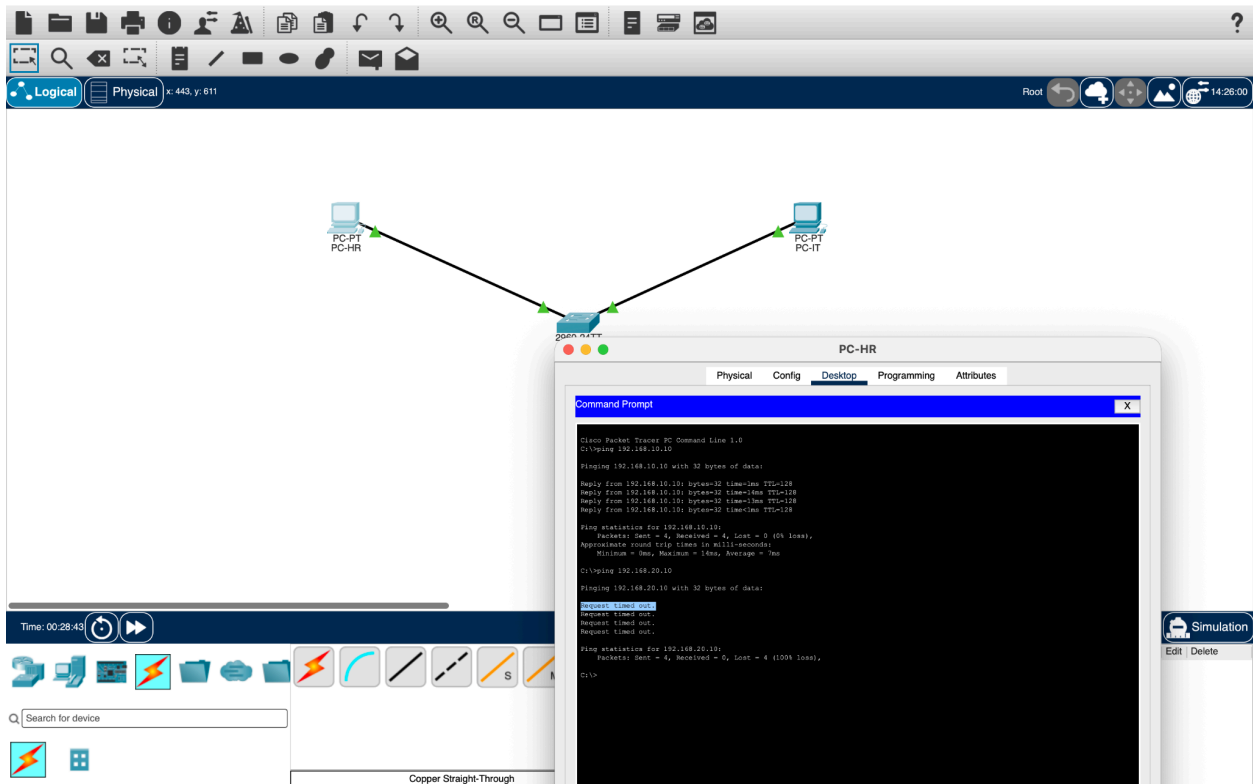
Purpose:

Demonstrates configuration of endpoint switch ports and department-based VLAN assignment.

Evidence 3 — VLAN Segmentation Testing

Screenshot:

PC-HR attempting communication with PC-IT.



Purpose:

Demonstrates successful network segmentation and validates that separate VLANs require routing for inter-network communication.

8. Skills Demonstrated

- Cisco Packet Tracer
- VLAN Configuration
- Switching Fundamentals
- Network Segmentation
- Access Port Configuration
- Network Troubleshooting
- Enterprise Network Concepts
- Layer 2 Networking

9. Next Steps

Future expansion:

- Configure inter-VLAN routing
- Add router connectivity
- Test communication between VLANs
- Document Layer 3 routing concepts